Extending QUIC with Multicast

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What we want to talk about today

- Real world use cases, the problem with IP multicast today and how QUIC fits in
- Multiple working proof of concept implementations
- Why the time might be right now to add multicast to the QUIC toolkit

What we don’t want to talk about today

- Technical discussions
- Merits of one implementation, proposal or solution over another
- The exact shape and level of the drafts required to make this work
- The inherent privacy and security trade-offs of multicast

→ Happy to discuss all of this offline or on the list

Use cases

- Multicast has many traditional use cases:
  - Live media
  - Software updates
- Multicast is actively being used on the internet today
  - Broadpeak Multicast ABR was used to deliver world cup games via multicast

The problem

- The delivery path is complicated, not very secure and requires custom hardware
- The solutions are also all proprietary
- Despite this, it’s still being used

How QUIC fits in

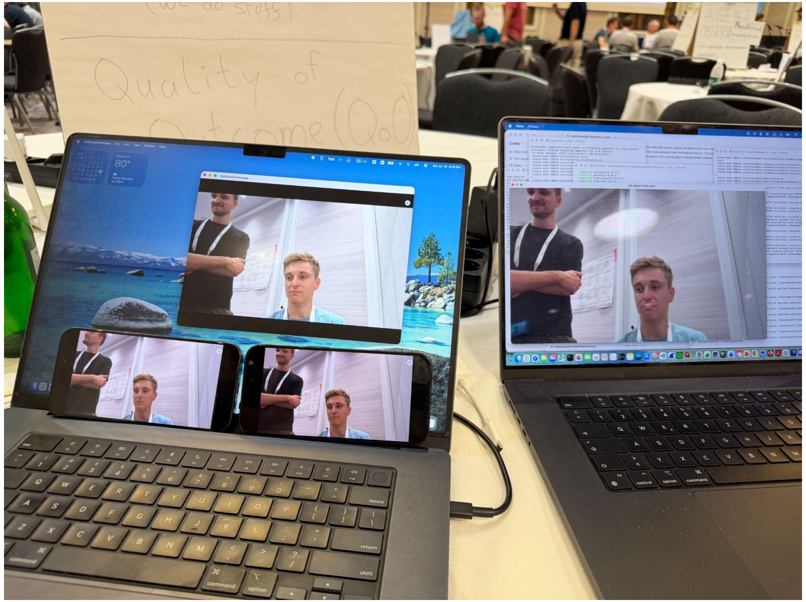
- QUIC runs on UDP, multicast comes very naturally
- QUIC gives path into browsers
- QUIC provides security and reliability
- QUIC allows end-to-end solution
- Extensions fit cleanly

We now have proof-of-concept implementations of 3 approaches

- Minimal multicast extension
- Flexicast
- MoQ over Multicast QUIC

Minimal Multicast QUIC: Lowering the bar to implement it

- Small set of modifications to QUIC (600 lines for the server)
  - 1 frame to announce the multicast tree and decryption key
  - Server sends DATAGRAMs on a new connection with an IP multicast address
  - Clients open a new socket and feed the content to the initial connection (other PNS)
- Implemented during this hackathon: 6 different QUIC stacks
- RTP stream application, working on laptops and iPhones



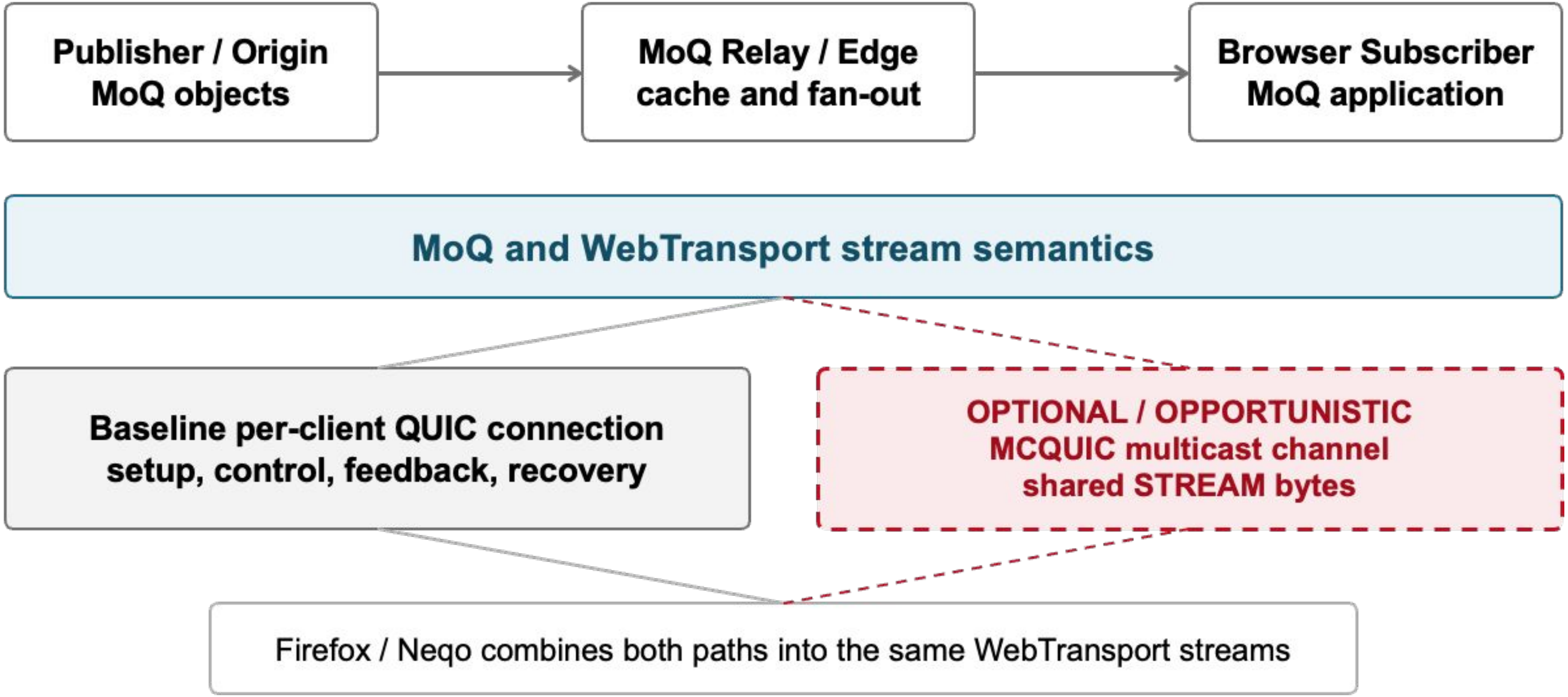
Flexicast: reusing Multipath QUIC semantics

- Reuse Multipath QUIC
  - Reliability, acknowledgments
  - Path management (fallback)
  - Congestion control
- Implementation in quiche
  - Scalable: up to 100 Gbps of multicast traffic
- Research paper about it
- draft-navarre-quic-flexicast

MoQ over Multicast QUIC

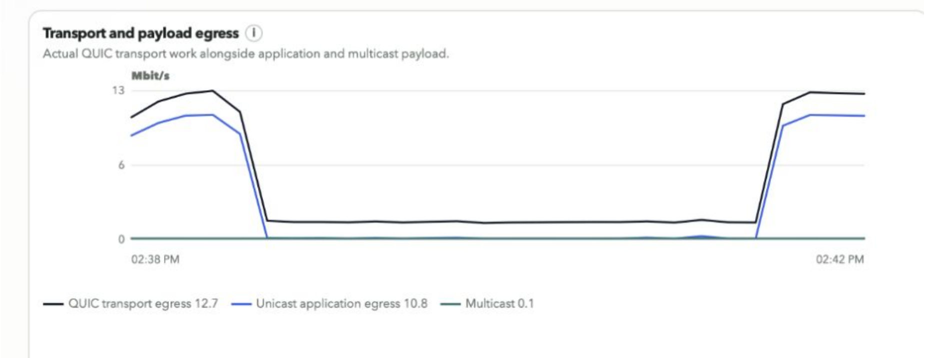
- draft-jholland-quic-multicast
- Uses MoQ architecture and a modified firefox/nightly to get live video stream over multicast in the browser
- MoQ relays emit multicast channels in addition to regular unicast connections
- Clients who support multicast will negotiate it and try to join the channel
- If they receive packets over multicast unicast transmission will stop
- This is using reliable STREAM frames over multicast
- Integrity frames ensure no third parties can inject traffic

MoQ over Multicast QUIC



MoQ over Multicast QUIC

- Seamless for clients, only if a client supports the extension and has a live multicast path it will receive the video via multicast otherwise over unicast
- This also means very low overhead for the server if multicast doesn't work, only a few additional frames sent at the start of the connection
- Our naive unoptimized implementation already shows a reduction of egress traffic of 80%
- Multicast path adds around 60ms of latency e2e, including integrity etc.



MoQ over Multicast QUIC

- You can try it today yourself on our demo website, extensions and modified browser are open source:
  - https://github.com/QUICast/firefox
  - https://live.quicast.de/

Why now

- We got more running code
- MoQ is a very attractive delivery path
- Multicast is getting used more and more for video delivery anyway:
    - Broadpeak
    - Aircast
    - Blockcast
- Multipath QUIC is done, so there is time and it could work as a base for Multicast QUIC

What we would like from the WG

- Discussion and feedback on multicast in QUIC in general
- People to take a look at the PoC and demos
- We are also looking for collaborators to maybe extend the demos or have an actual pilot deployment
    - We have public funding for this for the next year so we can heavily contribute and do the work here

Two existing, “complete” drafts for multicast in QUIC

- draft-jholland-quic-multicast
  - Rather complete, extends QUIC
  - Working PoC implementation with MoQ, WebTransport, support in the browser (Mozilla)
- draft-navarre-quic-flexicast
  - Focus on flexibility, multicast failures, scalability
  - Extends Multipath QUIC